

財政學 HW6

1. (Hyman 13.4) Suppose leisure is an inferior good for a worker. Set up this worker's indifference curves for money income and leisure, and derive the income and substitution effects of a tax-induced wage decline. Derive the compensated labor supply curve for this worker, and explain how it differs from the compensated supply curve of a worker for whom leisure is a normal good.

2. (Hyman 13.5) Empirical studies show that the market labor supply of prime-age males (between the ages of 25 and 55) in the United States is close to perfectly inelastic with respect to the labor compensation. If this is the case, explain why it does not imply that the excess burden of a tax on the labor income of prime-age males is necessarily zero. What does a perfectly inelastic market labor supply imply about the incidence of taxes on labor income? Why is there reason to believe that the overall market labor supply is not perfectly elastic when groups other than prime-age males are considered? What possible problems exist in empirical studies of the response to taxes on labor income that might make it difficult to estimate actual labor market responses?

3. (Rosen 18.5) According to Feldstein[2008b], “only about 10% to 20%” of a one-time tax rebate provided to taxpayers in 2008 was spent immediately. Using the life-cycle model of Figure 18.7, explain why this result could have been predicted.(Hint: Think about how a one-time rebate affects the endowment point and budget constraint in Figure 18.7. Contrast this to how a permanent rebate affects the endowment point and budget constraint.)

4. (Rosen 18.10) In an economy, the supply curve of labor, S , is given by

$$S = -100 + 200W_n$$

Where W_n is the after-tax wage rate. Assume that the before-tax wage rate is fixed at 10.

a. Write a formula for tax revenue as a function of the tax rate, and sketch function in a diagram with the tax rate on the horizontal axis and tax revenues on the vertical axis.

[Hint: Note that τ is the tax rate, and that tax revenue is the product of hours worked, the gross wage, and the tax rate.] Suppose that the government currently imposes a tax rate of 70 percent. What advice would you give it?

b. At what tax rate are tax revenues maximized in this economy?

5. (Rosen 19.1) Caterpillar Inc.'s CEO, Jim Owen, noted, "Sitting on a big wad of cash doesn't make any sense whatsoever for shareholders [because] it tends to promote bad practice among management.... You've got more [cash] than you know what to do with, and you think you're so damn good you can buy anything and make it better" [Brat and Gruley, 2007]. If this view is correct, what does it imply about the efficiency consequences of the corporate tax treatment of dividends versus retained earnings?

6. (Rosen 19.11) The ABC corporation is contemplating purchasing a new computer system that would yield a before-tax return of 30 percent. The system would depreciate at a rate of 1 percent a year. The after-tax interest rate is 8 percent, the corporation tax rate is 35 percent, and a typical shareholder of ABC has a marginal tax rate of 30 percent. Assume for simplicity that there are no depreciation allowances or investment tax credits. Do you expect ABC to buy the new computer system? Explain your answer. [Hint Use Equation (19.4)]